Chapter 8

Basic Definitions

8.1 Undirected Graphs and Directed Graphs

Definition: A ***graph*** is a triple $G = (\ V, E, \psi\ )$ in which V and E are two disjoint finite sets such that $V \neq \emptyset$ and ψ is a function such that, either $\psi : E \to \{\{u, v\} \mid u, v \in V\}$ or $\psi : E \to V \times V$. In the former case, G is an ***undirected graph***, while in the latter case, G is a ***directed graph***. Each element of V is called a ***vertex*** of G, and each element of E is called an ***edge*** of G. The set V is called the ***vertex set*** of G while the set E is called the ***edge set*** of G. The function ψ is called the ***incidence function*** of G. The ***order*** of G is the number of vertices in G (i.e. $|V|$) and the ***size*** of G is the number of edges in G (i.e. $|E|$). For notational convenience, a graph is frequently represented by solely a letter, say G, and the vertex set, edge set and the incidence function of the graph are represented by V_G, E_G and ψ_G, respectively.

In undirected graphs, edges are mapped onto *unordered* pairs of vertices, while in directed graphs, edges are mapped onto *ordered* pairs of vertices. Note that for undirected graphs, an edge e can be mapped onto $\{u, v\}$ such that $u = v$. In which case, the edge e is actually mapped onto a singleton $\{u\}$ instead of an unordered pair. However, we shall write $\psi(e) = \{u, u\}$ instead of $\psi(e) = \{u\}$.

Example 8.1.1 Let $G = (V, E, \psi)$ where

$V = \{v_1, v_2, v_3, v_4, v_5\}$,

$E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8\}$, and

$\psi : E \to \{\{u, v\} \mid u, v \in V\}$ such that

$\psi(e_1) = \{v_1, v_2\}$, $\psi(e_2) = \{v_2, v_3\}$, $\psi(e_3) = \{v_3, v_3\}$,

$\psi(e_4) = \{v_3, v_4\}$, $\psi(e_5) = \{v_2, v_4\}$, $\psi(e_6) = \{v_4, v_5\}$,

$\psi(e_7) = \{v_2, v_5\}$, $\psi(e_8) = \{v_2, v_5\}$.

Then, according to the above definition, G is an undirected graph of order 5 and size 8. Note that we write $\psi(e_3) = \{v_3, v_3\}$ instead of $\psi(e_3) = \{v_3\}$. ■

Example 8.1.2 Let $G' = (V', E', \psi')$ where

$V' = \{a, b, c, d\}$,

$E' = \{e_1, e_2, e_3, e_4\}$ and

$\psi' = \{(e_1, (a, b)), (e_2, (b, a)), (e_3, (b, c)), (e_4, (c, d))\}$

Then, according to the above definition, G' is a directed graph of order 4 and size 4. ■

8.2 Graphical Representation of Graphs

Graphs, as their name suggests, have a convenient graphical representation. Given a graph $G = (V, E, \psi)$. For each vertex $v \in V$, the vertex v is represented by a dot with a label v (Figure 8.2.1). For each edge $e \in E$, if G is an undirected graph and $\psi(e) = \{u, v\}$, the edge e is represented by a line segment (not necessarily straight) joining the two dots representing vertices u and v (Figure 8.2.2). If G is a directed graph and $\psi(e) = (u, v)$, the edge e is represented by an arrow with the two dots representing the vertices u and v as its tail and head, respectively (Figure 8.2.3). In either case, the edge or arrow is labelled e.

Figure 8.2.1 Representation of vertex a



Figure 8.2.2 Representation of an undirected edge e where $\psi(e)=\{a,b\}$

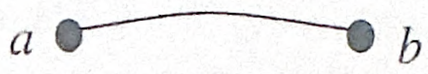
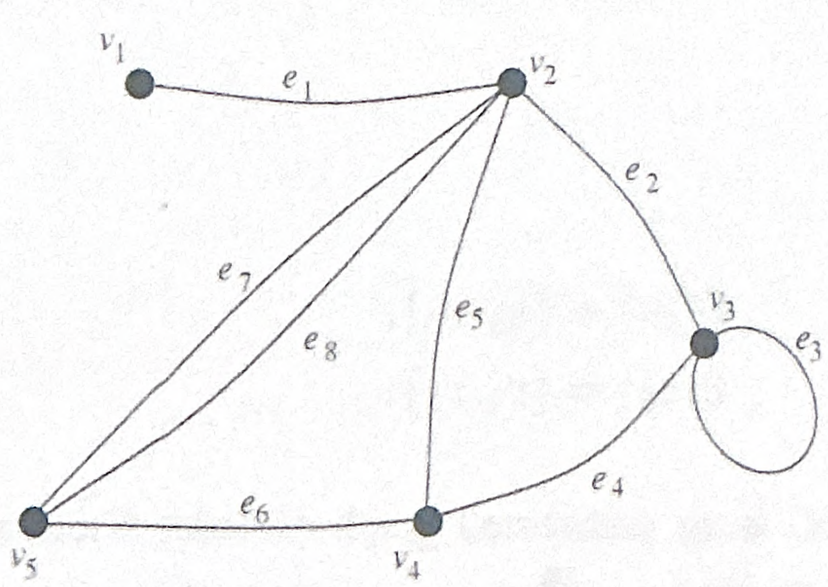
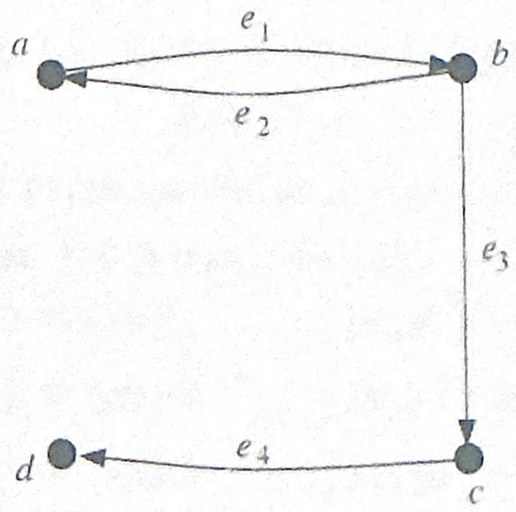


Figure 8.2.3 Representation of a directed edge$e$ where $\psi(e)=(a,b)$



Example 8.2.3 The following are graphical representations of the two graphs defined in Examples 8.1.1 and 8.1.2. ■





Example 8.2.4 Let $G = (V, E, \psi)$ where

$V = \{a, b, c, d, e, f, g, h\}$,

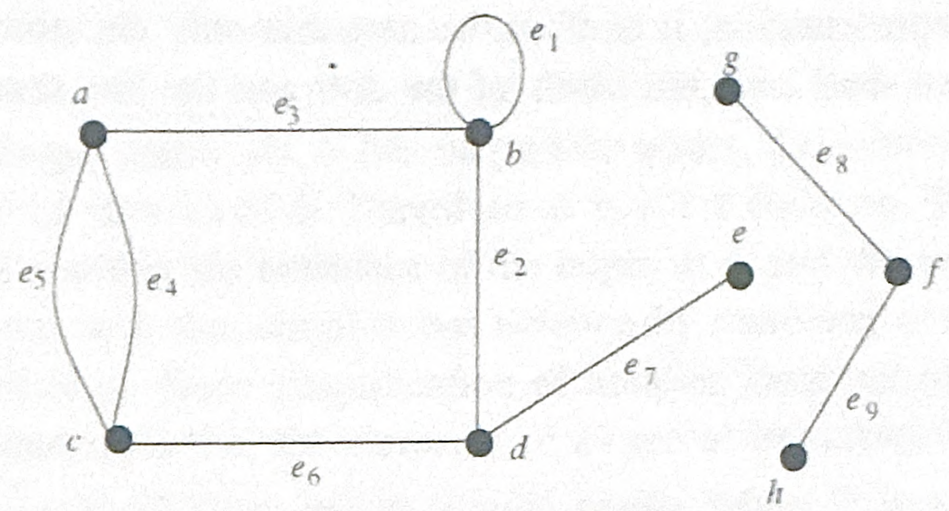
$E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7, e_8, e_9\}$ and

$\psi : E \to \{\{u, v\} \mid u, v \in V\}$ be such that:

$\psi(e_1) = \{b, b\}$, $\psi(e_2) = \{b, d\}$,
$\psi(e_3) = \{a, b\}$, $\psi(e_4) = \{a, c\}$,
$\psi(e_5) = \{a, c\}$, $\psi(e_6) = \{c, d\}$,
$\psi(e_7) = \{d, e\}$, $\psi(e_8) = \{f, g\}$
$\psi(e_9) = \{h, f\}$.

A graphical representation of G is shown in Figure 4.

Figure 4



Example 8.2.5 Let $G = (V, E, \psi)$ where

$V = \{a,b,c,d\}$,

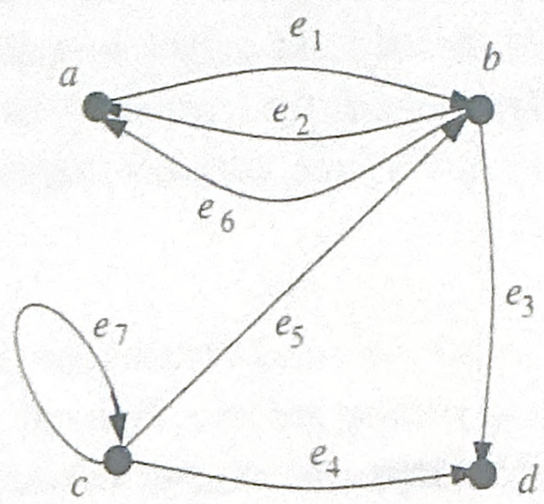
$E = \{e_1, e_2, e_3, e_4, e_5, e_6, e_7\}$ and

$\psi : E \to V \times V$ such that;

$\psi(e_1) = (a, b)$, $\psi(e_2) = (b, a)$,
$\psi(e_3) = (b, d)$, $\psi(e_4) = (c, d)$,
$\psi(e_5) = (c, b)$, $\psi(e_6) = (b, a)$,
$\psi(e_7) = (c, c)$

A graphical representation of G is shown in Figure 5.

Figure 5



Since in drawing out a graphical representation, we are free to deploy the dots representing the vertices anywhere on the plane and are free to draw the line segments representing the edges in any way we like, it is obvious that corresponding to every graph, there is more than one graphical representations.

From the above examples, it should be obvious that while the set notation definition provides a formal way of defining graphs, the graphical representation provides an intuitively clearer way of defining them. For instance, in Example 2, the graphical representation of the graph G clearly shows that there is a chain of edges in the graph connecting vertices g and h, whereas there is no chain of edges connecting vertices a and g. These properties of G are not obvious in the set notation definition of G. In most cases, while the set notation definition of a graph conveys very little information about the properties of the graph, its graphical representation normally displays a great deal of information about the properties of the graph. It is for this reason that we shall frequently define graphs using their graphical representations and shall refer to the latter as graphs themselves in subsequent discussion.

In using graphical representation, it is often the case that only the structural properties of the graph matter. In those cases, we shall omit the labels of the dots and the line segments in the representation.

8.3 Adjacency

Definition: Let $G = (V, E, \psi)$ be a graph and $e \in E$. If $\psi(e) = \{a, b\}$ or $\psi(e) = (a, b)$, then a and b are the ***end-vertices*** of e. In the case where $\psi(e) = (a, b)$, a is the ***tail*** of e while b is the ***head*** of e. In either case, we say that edge e ***joins*** the vertices a and b, edge e is ***incident upon*** vertices a and b, and vertex a (b, respectively) is ***adjacent to*** vertex b (a, respectively). Two edges are ***adjacent edges*** if they share a common end-vertex. Note that when G is an undirected graph, vertex v is an end-vertex of edge e if and only if $v \in \psi(e)$.

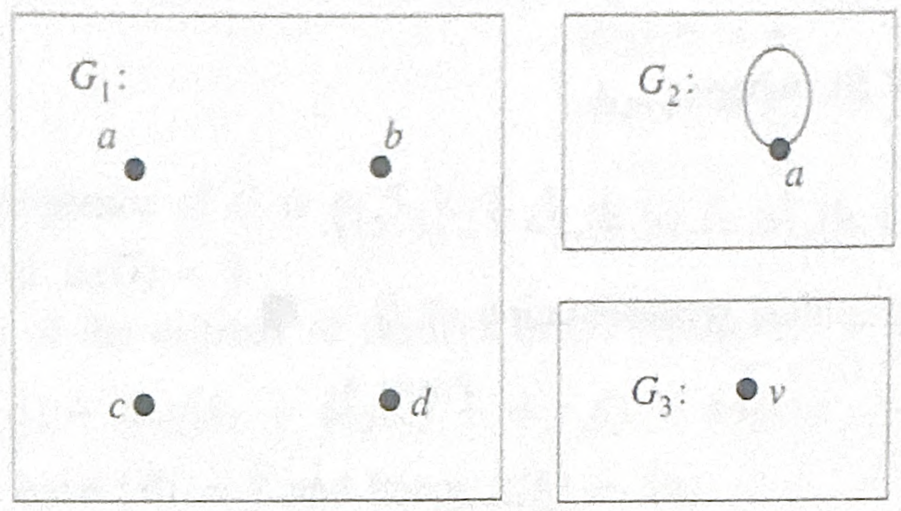
In the undirected graph in Example 8.2.3, vertices v_1, v_2 are the end-vertices of edge e_1; vertex v_3 is the only end-vertex of edge e_3, and vertices v_2, v_5 are the end-vertices of edges e_7 and e_8. Edge e_1 joins vertices v_1 and v_2, or equivalently, e_1 is incident upon vertices v_1 and v_2. Moreover, vertex v_1 is adjacent to vertex v_2. The edges e_4 and e_5 are adjacent edges, so are the edges e_2 and e_3.

In the directed graph in Example 5, vertices a and b are the end-vertices of both the edges e_1 and e_2. However, in edge e_1, vertex a is the tail and vertex b is the head whereas in edge e_2, vertex a is the head and vertex b is the tail. Edge e_3 joins the vertices b and d, or equivalently, edge e_3 is incident upon vertices b and d. The edges e_3 and e_4 are adjacent edges.

Definition: A ***self-loop*** is an edge whose two end-vertices are identical. Two edges are ***parallel edges*** if they have the same end-vertices and if they are in a directed graph, they also have the same tail and the same head. Graphs having no self-loops and no parallel edges are called ***simple graphs***. A graph without any edge is called a ***null graph***. A null graph which has exactly one vertex is called a ***trivial graph***. A graph consisting of one vertex and one self-loop is called a ***loop graph***.

In Figure 6, G_1 is a null graph of order 4, G_2 is a loop graph, and G_3 is a trivial graph.

Figure 6



Let $G=(V, E,\psi)$ be a simple graph. As G has no parallel edges, the incidence function ψ maps every edge in G to a distinct pair of vertices of G. Therefore, ψ is a 1-1 function. Since the elements of E are of no significant other than marking the existence of the edges in G and the vertex pairs in $Ran(\psi)$ could accomplish the same task, we shall thus simplify our notation by removing ψ from (V,E,ψ) and replacing E by the range of ψ, i.e. $Ran(\psi)$. Such simplification of notation does not result in loss of information of G, unless the labels of the edges (i.e. the elements of E) are of importance which is rarely the case.

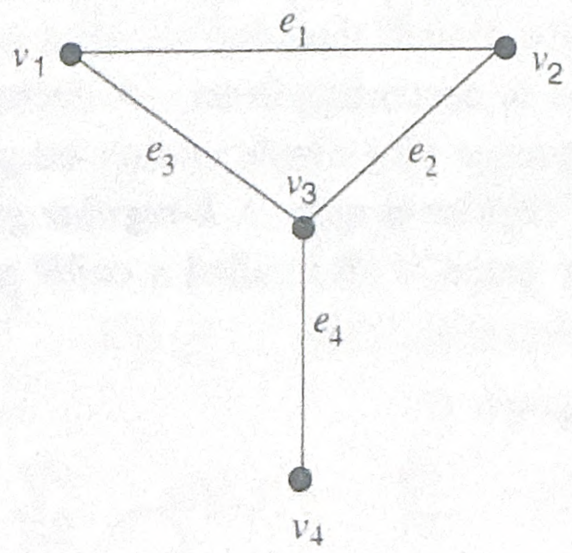
Henceforth, we shall use $G=(V,E)$ to denote simple graphs where V is a set of vertices and E is a set of ordered or unordered pairs of vertices representing the edges.

Example 8.3.6 Consider the simple graph: $G=(V,E,\psi)$, where

$$V = \{v_1, v_2, v_3, v_4\},$$
$$E = \{e_1, e_2, e_3, e_4\}, \text{ and}$$
$$\psi = \{(e_1, \{v_1, v_2\}), (e_2, \{v_2, v_3\}), (e_3, \{v_1, v_3\}), (e_4, \{v_3, v_4\})\}.$$

The following is a graphical representation of G.

Figure 7



Using the simplified notation for simple graphs, the graph G is defined as: $G=(V,E)$, where

$$V = \{v_1, v_2, v_3, v_4\}, \text{ and}$$
$$E = \{\{v_1, v_2\}, \{v_2, v_3\}, \{v_1, v_3\}, \{v_3, v_4\}\}.$$

This definition of G gives rise to the same graphical representation except the labels of the edges are removed. ■

Example 8.3.7 The null graph G_1 in Figure 6 can be defined as follows using the simplified notation.

G=(V,E), where $V = \{a, b, c, d\}$ and $E = \emptyset$. ■

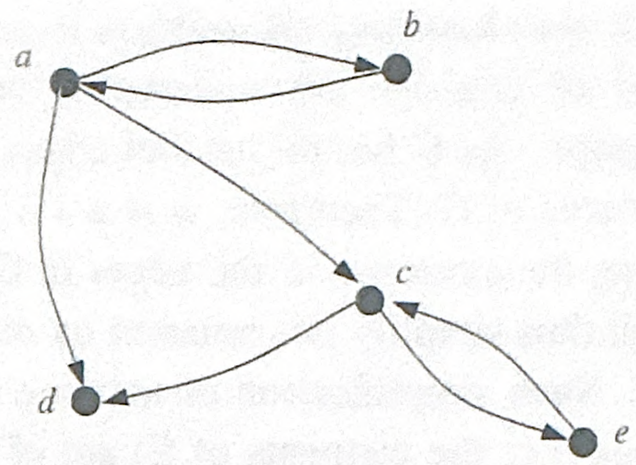
Example 8.3.8 Let G=(V,E), where

$V = \{a, b, c, d, e\}$.

$E = \{(a, b), (b, a), (a, d), (c, e), (c, d), (a, c), (e, c)\}$.

The following is a graphical representation of G. ■

Figure 8



8.4 Degree Sequence

Definition: Let G=(V, E,ψ) be an undirected graph. The number of edges incident upon a vertex v in G, with self-loops counted twice, is called the ***degree*** (***valency***) of v in G and is denoted by $deg_G(v)$ (The subscript G is omitted when no confusion could occur). More formally,

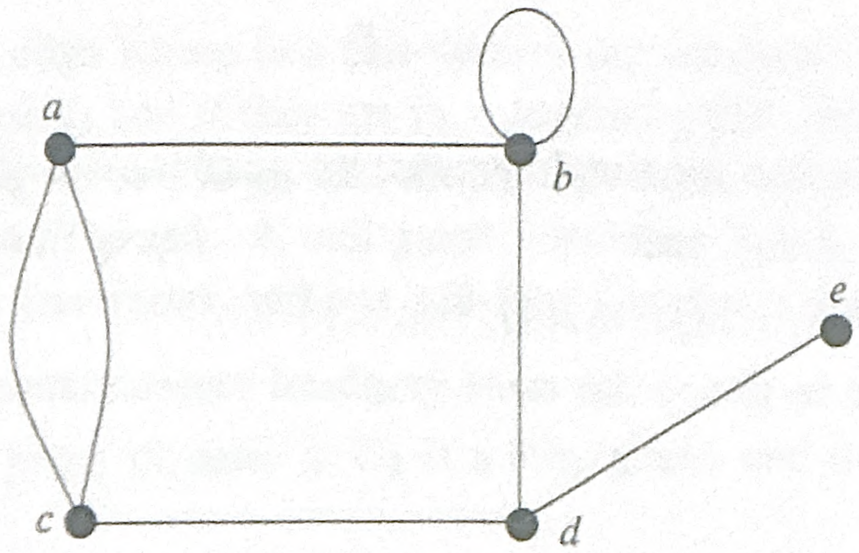
$$deg_G(v) = |\{e \in E \mid v \in \psi(e) \wedge \psi(e) \neq \{v, v\}\}| + 2|\{e \in E \mid \psi(e) = \{v, v\}\}|.$$

Note that if G is a simple graph, then

$$deg_G(v) = |\{e \in E \mid v \in \psi(e)\}|, \text{ as } \{e \in E \mid \psi(e) = \{v, v\}\} = \emptyset.$$

The *smallest* and *largest* degree of vertices in G are denoted by $\delta(G)$ and $\Delta(G)$, respectively. Let $V = \{v_1, v_2, v_3, \cdots, v_p\}$. The ***degree sequence*** of G=(V,E,ψ) is the sequence $deg(v_1), deg(v_2), deg(v_3), \cdots, deg(v_p)$ such that $deg(v_i) \geq deg(v_{i+1}), 1 \leq i < p$ (i.e. the sequence of degrees of vertices of G is listed in decreasing order). A decreasing sequence of integers is a ***graphical sequence*** if it is the degree sequence of a simple undirected graph. A graph in which all the vertices are of the same degree is called a ***regular graph***. A ***k-regular graph*** is a regular graph in which every vertex is of degree k. A 3–regular graph is also called a ***cubic graph***. A vertex v is an ***isolated vertex*** if $deg(v)$=0 and is a ***pendant*** if $deg(v)$=1.

Example 8.4.9 In the following graph G:



$deg(b) = 4$

$deg(c) = 3$

$deg(a) = 3$ $\qquad$ $deg(d) = 3$

$deg(e) = 1$

Therefore, the degree sequence of G is 4, 3, 3, 3, 1.

Moreover, $\delta(G) = 1$ and $\Delta(G) = 4$.

Now, consider the sum of the degrees of vertices in G. We have:

$$deg(a) + deg(b) + deg(c) + deg(d) + deg(e) = 14$$

On the other hand, we have $|E| = 7$ and hence $2|E| = 14$.

Therefore, the sum of the degrees of vertices in G is twice the size of G. The following theorem shows that this equality holds true in general. ■

Theorem 8.4.1: *Let $G=(V, E, \psi)$ be an undirected graph. Then*

$$\sum_{v \in V} deg(v) = 2|E|.$$

Proof: If $|E| = 0$, then $2|E| = 0$ and $deg(v) = 0, \forall v \in V$.

So $\sum_{v \in V} deg(v) = 0$, implying $\sum_{v \in V} deg(v) = 2|E|$.

Suppose $|E| > 0$. By the definition of graph, $|V| > 0$.

Let $V = \{v_1, v_2, \cdots, v_p\}$ and $E = \{e_1, e_2, \cdots, e_q\}$, where $|V| = p$ and $|E| = q$.

Construct a $p \times q$ matrix M as follows:

$$M[i,j] = \begin{cases} 0 & if\ v_i \notin \psi(e_j) \\ 1 & if\ v_i \in \psi(e_j) \wedge \psi(e_j) \neq \{v_i, v_i\} \\ 2 & if\ \psi(e_j) = \{v_i, v_i\} \end{cases}$$

i.e $M[i,j]$ is the number of times (0,1 or 2) edge e_j is incident upon vertex v_i. (Figure 8.4.9)

Then the sum of all the entries in row i is precisely $deg(v_i)$. Specifically,

$$\forall i, 1 \leq i \leq p, \sum_{j=1}^{q} M[i,j] = \sum_{\substack{j=1 \\ M[i,j] \neq 0}}^{q} M[i,j] + \sum_{\substack{j=1 \\ M[i,j]=0}}^{q} M[i,j] \qquad (M[i,j] \in \{0,1\}, 1 \leq j \leq q)$$

$$= \sum_{\substack{j=1 \\ M[i,j] \neq 0}}^{q} M[i,j] \qquad \left(\sum_{\substack{j=1 \\ M[i,j]=0}}^{q} M[i,j] = 0 \right)$$

$$= \sum_{\substack{j=1 \\ v_i \in \psi(e_j)}}^{q} M[i,j] \qquad (M[i,j] \neq 0 \Leftrightarrow v_i \in \psi(e_j))$$

$$= \sum_{\substack{j=1 \\ v_i \in \psi(e_j) \\ \psi(e_j) \neq \{v_i, v_i\}}}^{q} M[i,j] + \sum_{\substack{j=1 \\ v_i \in \psi(e_j) \\ \psi(e_j) = \{v_i, v_i\}}}^{q} M[i,j] \qquad \begin{array}{r} (v_i \in \psi(e_j) \Leftrightarrow v_i \in \psi(e_j) \wedge (\psi(e_j) \neq \{v_i, v_i\}) \\ \wedge v_i \in \psi(e_j) \wedge (\psi(e_j) = \{v_i, v_i\})) \end{array}$$

$$= \sum_{\substack{j=1 \\ v_i \in \psi(e_j) \\ \psi(e_j) \neq \{v_i, v_i\}}}^{q} 1 + \sum_{\substack{j=1 \\ v_i \in \psi(e_j) \\ \psi(e_j) = \{v_i, v_i\}}}^{q} 2 \qquad \text{(Definition of } M[i,j])$$

$$= |\{e \mid v_i \in \psi(e) \wedge \psi(e) \neq \{v_i, v_i\}\}| + 2 \cdot |\{e \mid \psi(e) = \{v_i, v_i\}\}|$$

$$= deg(v_i) \qquad \text{(Definition of } deg(v_i)) \cdots \text{(I)}$$

Therefore, $\sum\limits_{\substack{1\le i\le p\\ 1\le j\le q}} M[i,j] = \sum\limits_{i=1}^{p}\sum\limits_{j=1}^{q} M[i,j]$ (+ is commutative and associative)

$$= \sum_{i=1}^{p} deg(v_i) \qquad ((\mathrm{I}),\ \mathrm{Sub}_{=})$$

$$= \sum_{v\in V} deg(v) \qquad (\text{Definition of } V)\ \cdots (\mathrm{II})$$

	e_1	e_2	...	e_j	...	e_q	*Row Sums*
v_1	$M[1,1]$	$M[1,2]$	...	$M[1,j]$	...	$M[1,q]$	$deg(v_1)$
v_2	$M[2,1]$	$M[2,2]$	...	$M[2,j]$	...	$M[2,q]$	$deg(v_2)$
⋮	⋮	⋮		⋮		⋮	
v_i	$M[i,1]$	$M[i,2]$	...	$M[i,j]$	...	$M[i,q]$	$deg(v_i)$
⋮	⋮	⋮		⋮		⋮	
v_p	$M[p,1]$	$M[p,2]$	...	$M[p,j]$	...	$M[p,q]$	$deg(v_p)$
Column Sums	2	2		2		2	

Figure 8.4.9 The Matrix M

On the other hand, the sum of all the entries in each column is 2 (because every edge has two end-vertices). Specifically, $\forall j, 1 \le j \le q$, let $\psi(e_j) = \{v_h, v_k\}$,

$$\sum_{i=1}^{p} M[i,j] = \sum_{\substack{i=1\\ M[i,j]\neq 0}}^{p} M[i,j] + \sum_{\substack{i=1\\ M[i,j]=0}}^{p} M[i,j] \qquad (M[i,j]\in\{0,1\}, 1\le i\le p)$$

$$= \sum_{\substack{i=1\\ M[i,j]\neq 0}}^{p} M[i,j] \qquad \left(\sum_{\substack{i=1\\ M[i,j]=0}}^{p} M[i,j] = 0\right)$$

$$= \sum_{\substack{i=1\\ v_i\in\psi(e_j)}}^{p} M[i,j] \qquad (M[i,j]\neq 0 \Leftrightarrow v_i\in\psi(e_j))$$

$$= \sum_{\substack{i=1\\ v_i\in\{v_h,v_k\}}}^{p} M[i,j] \qquad (\psi(e_j) = \{v_h, v_k\})$$

$$= \begin{cases} M[h,j] + M[k,j] & if\ h\neq k \\ M[h,j] & if\ h = k \end{cases}$$

$$= \begin{cases} 1+1 & if\ \psi(e_j) = \{v_h, v_k\} \\ 2 & if\ \psi(e_j) = \{v_h, v_h\} \end{cases} \qquad (\text{Definition of } M[i,j])$$

$$= 2 \quad \cdots (\mathrm{III})$$

Therefore, $\sum\limits_{\substack{1\le i\le p\\ 1\le j\le q}} M[i,j] = \sum\limits_{j=1}^{q}\sum\limits_{i=1}^{p} M[i,j]$ (+ is commutative and associative)

$$= \sum_{j=1}^{q} 2 \qquad ((\mathrm{III}),\ \mathrm{Sub}_{=})$$

$$= 2|E| \qquad \text{(Definition of } E\text{)} \cdots \text{(IV)}$$

Hence, $\sum_{v \in V} deg(v) = 2|E|$. ((II),(IV), $\text{Sub}_=$) ■

Theorem 8.4.2: *The number of vertices of odd degree in an undirected graph G=(V,E,ψ) is even.*

Proof: Let $V = \{w_1, w_2, \cdots, w_p\}$. Let $V_e = \{u_1, u_2, \cdots, u_m\}$ and $V_o = \{v_1, v_2, \cdots, v_n\}$ be the set of vertices of even and odd degree, respectively, in G. We are to prove that n is even.

Since every vertex is of either even or odd degree, therefore $V_e \cup V_o = V$ and $V_e \cap V_o = \emptyset$.

It follows that $\sum_{v \in V} deg(v) = \sum_{v \in V_e \cup V_o} deg(v)$

$$= \sum_{v \in V_e} deg(v) + \sum_{v \in V_o} deg(v). \cdots\cdots \text{(I)}$$

By Theorem 8.4.1, $\sum_{v \in V} deg(v) = 2|E|$.

We thus have $\sum_{v \in V_e} deg(v) + \sum_{v \in V_o} deg(v) = 2|E|$ ((I), $\text{Sub}_=$)

$$\Rightarrow \sum_{i=1}^{m} deg(u_i) + \sum_{i=1}^{n} deg(v_i) = 2|E| \qquad \text{(Definition of } V_e \text{ and } V_o\text{)}$$

$$\Rightarrow \sum_{i=1}^{n} deg(v_i) = 2|E| - \sum_{i=1}^{m} deg(u_i). \cdots\cdots \text{(II)}$$

Since $deg(u_i), 1 \le i \le m$, is even, $(\exists h \in \mathbf{N})(deg(u_i) = 2h), 1 \le i \le m$ (Def of even integer)

$\Rightarrow h_i \in \mathbf{N} \wedge (deg(u_i) = 2h_i), 1 \le i \le m$ (EI)

$\Rightarrow h_i \in \mathbf{N}$ and $deg(u_i) = 2h_i, 1 \le i \le m$ (E9. I2) $\cdots$ (III)

Since $deg(v_i), 1 \le i \le n$, is odd, $(\exists k \in \mathbf{N})(deg(v_i) = 2k + 1), 1 \le i \le n$ (Def of odd integer)

$\Rightarrow k_i \in \mathbf{N} \wedge (deg(v_i) = 2k_i + 1), 1 \le i \le n$ (EI)

$\Rightarrow k_i \in \mathbf{N}$ and $deg(v_i) = 2k_i + 1, 1 \le i \le n$ (E9. I2) $\cdots$ (IV)

It follows that $\sum_{i=1}^{n} (2k_i + 1) = 2|E| - \sum_{i=1}^{m} 2h_i$ ((II),(III),(IV), $\text{Sub}_=$)

$$\Rightarrow \sum_{i=1}^{n} 2k_i + \sum_{i=1}^{n} 1 = 2|E| - \sum_{i=1}^{m} 2h_i$$

$$\Rightarrow \sum_{i=1}^{n} 2k_i + n = 2|E| - \sum_{i=1}^{m} 2h_i$$

$$\Rightarrow n = 2\left(|E| - \sum_{i=1}^{m} h_i - \sum_{i=1}^{n} k_i\right), \text{ where } |E| - \sum_{i=1}^{m} h_i - \sum_{i=1}^{n} k_i \in \mathbf{N}$$

$\Rightarrow (\exists K \in \mathbf{N})(n = 2K)$ (EQ)

$\Rightarrow n$ is even. ■

Using Theorem 8.4.1, an upper bound on the size of simple undirected graphs is easily derived.

Corollary 8.4.3: *Let $G = (V, E)$ be a simple undirected graph. Then $|E| \le \frac{|V|(|V|-1)}{2}$.*

Proof: Let $v \in V$.

Since G is simple, between v and every other vertex, there is at most one edge joining them.

This implies there can be at most $|V| - 1$ edges incident upon v.

In other words, $deg(v) \le |V| - 1, \forall v \in V$ (I).

By Theorem 8.4.1, we have:

$2|E| = \sum_{v \in V} deg(v) \le \sum_{v \in V} (|V| - 1)$ ($\because$ (I))

$$= |V|(|V| - 1).$$

Hence, $|E| \leq \frac{|V|(|V|-1)}{2}$ ■

8.5 Subgraphs

Definition: Let $G=(V,E,\psi)$ and $G'=(V',E',\psi')$ be two graphs. G is ***identical to*** G', denoted by $G = G'$, if $V = V'$, $E = E'$ and $\psi=\psi'$. G' is a ***subgraph*** of G and G is a ***supergraph*** of G', denoted by $G' \sqsubseteq G$, if $V' \subseteq V$, $E' \subseteq E$ and $\psi' \subseteq \psi$. G' is a ***proper subgraph*** of G, denoted by $G' \sqsubset G$, if $G' \sqsubseteq G$ and $G \neq G'$. G' is a ***spanning subgraph*** of G if $V'=V$ and G' is a subgraph of G. The ***underlying simple graph*** of G is the simple graph resulting from G after all the self-loops are removed, and for every pair of adjacent vertices having parallel edges joining them, all but one of the edges joining them are removed.

In Figure 10, a graph G and its underlying simple graph are displayed. In Figure 11, five subgraphs of graph G are shown, all of which, except G_3, are proper subgraphs of G. G_2 and G_3 are spanning subgraphs of G. Note that G_1 is also a subgraph of G_2, and G_5 is also a subgraph of G_2 and G_4.

Figure 10 A graph G and its underlying simple graph.

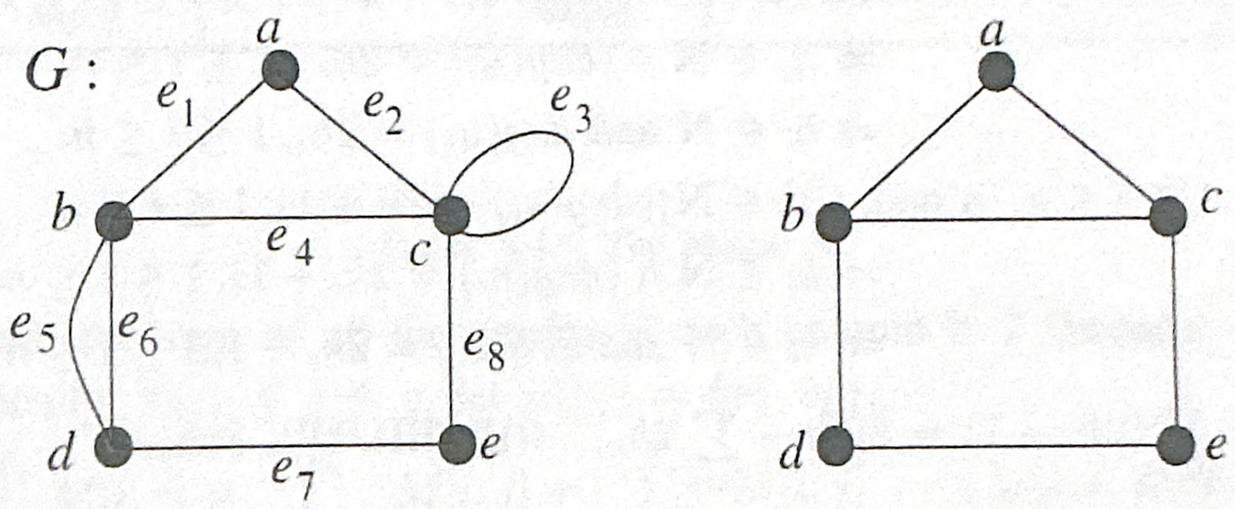
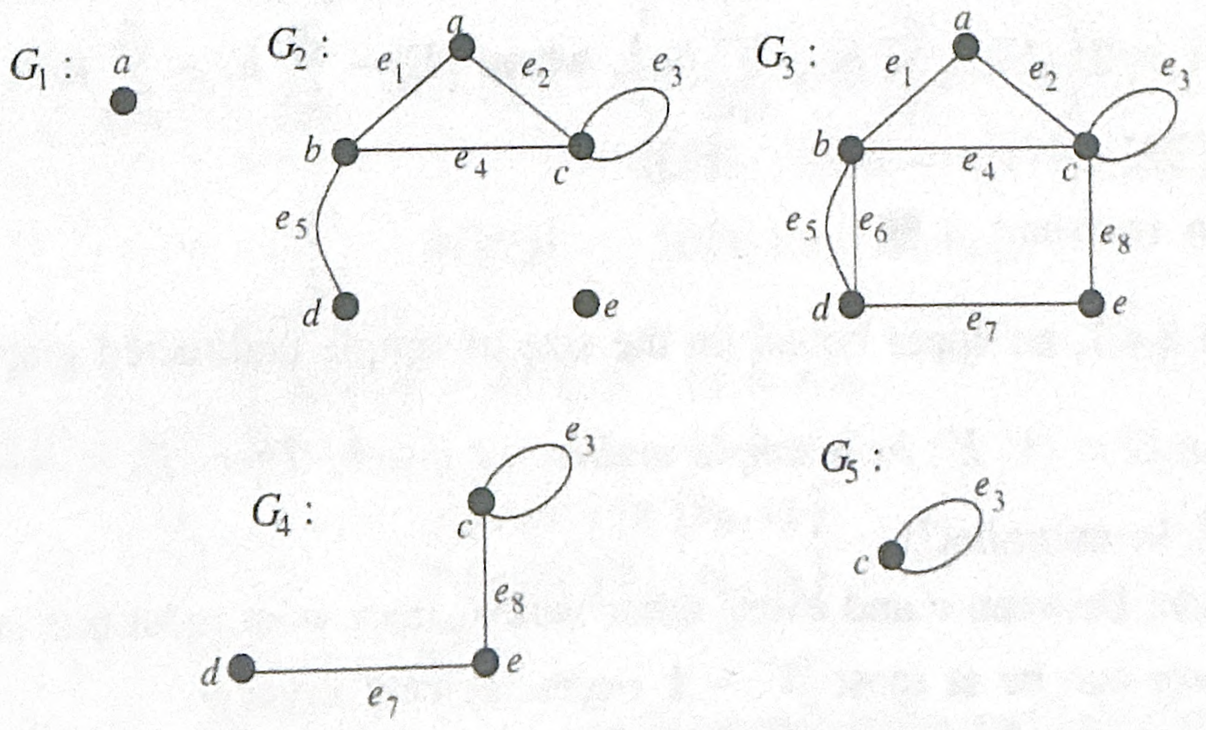


Figure 11 Some subgraphs of G.



Lemma 8.5.4: *The relation* $\sqsubseteq$ *is a partial order.*

Proof: (left as an exercise). ■

Lemma 8.5.5: *The relation $\sqsubset$ is a strict order.*

Proof: (left as an exercise). ■

Definition: Let $G=(V, E, \psi)$ be a graph. Let $U \subseteq V$ such that $U \neq \emptyset$. The subgraph of G ***induced by*** U, denoted by $\langle U \rangle_G$, is the largest subgraph of G with vertex set U. i.e. $\langle U \rangle_G \in \mathcal{G}_U \wedge (\forall H \in \mathcal{G}_U)(H \sqsubseteq \langle U \rangle_G)$, where $\mathcal{G}_U$ is the set of all subgraphs of G with vertex set U. (The subscript G can be omitted when no confusion could occur.)

In other words, every subgraph of G with vertex set U is a subgraph of $\langle U \rangle_G$. As a result, the graph $\langle U \rangle_G$ consists of the vertex set U and all the edges of G whose end-vertices are in U. Note that by Lemma 5.4.3, $\langle U \rangle_G$ is unique.

Example 8.5.10 Consider the following graph $G = (V, E)$ where $V = \{a, b, c, d, e, f, g\}$. Let $U = \{b, c, e\}$. A number of subgraphs of G with vertex set V are displayed in Figure 12, among which, the subgraph H_4 is the subgraph induced by U.

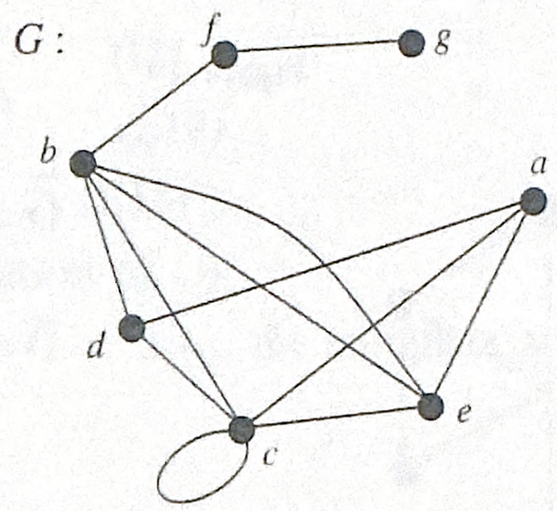
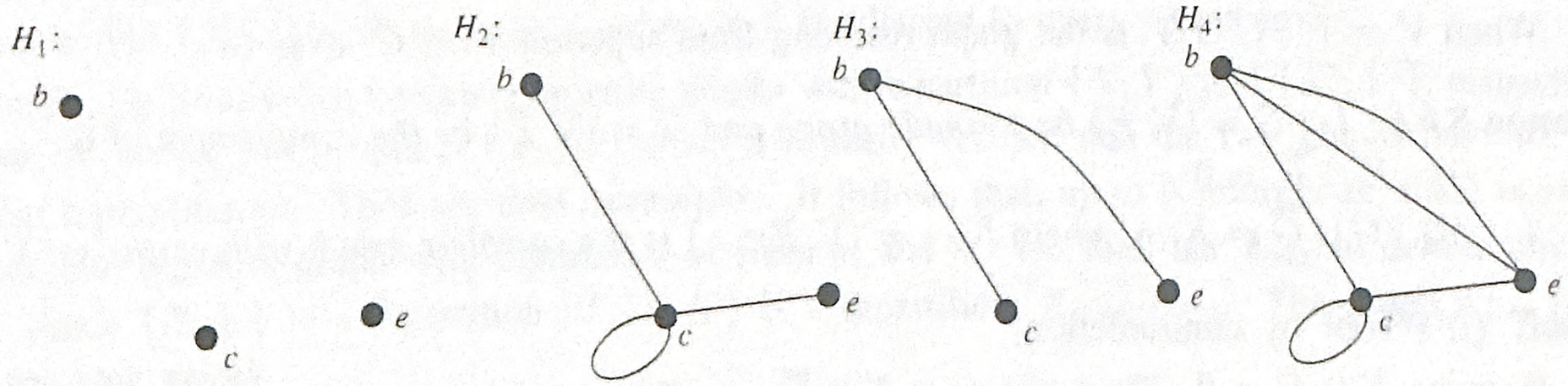


Figure 12 H_4 is the subgraph of G induced by $U = \{b, c, e\}$.

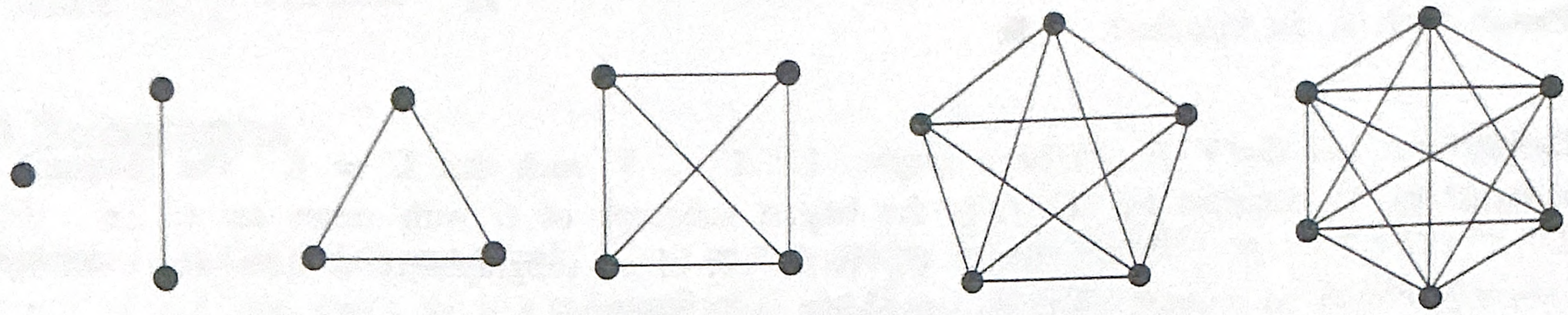


8.6 Complete Graphs

Definition: A ***complete*** graph is a simple undirected graph $G=(V,E)$ in which $\{u, v\} \in E, \forall u, v \in V, u \neq v$.

So, a complete graph is a simple graph in which every two distinct vertices are adjacent. As a direct consequence, in a complete graph of order p, $deg(v) = p - 1$ for every vertex v. In Figure 13, complete graphs of orders 1, 2, 3, 4, 5 and 6, respectively, are displayed.

Figure 13 Complete graphs



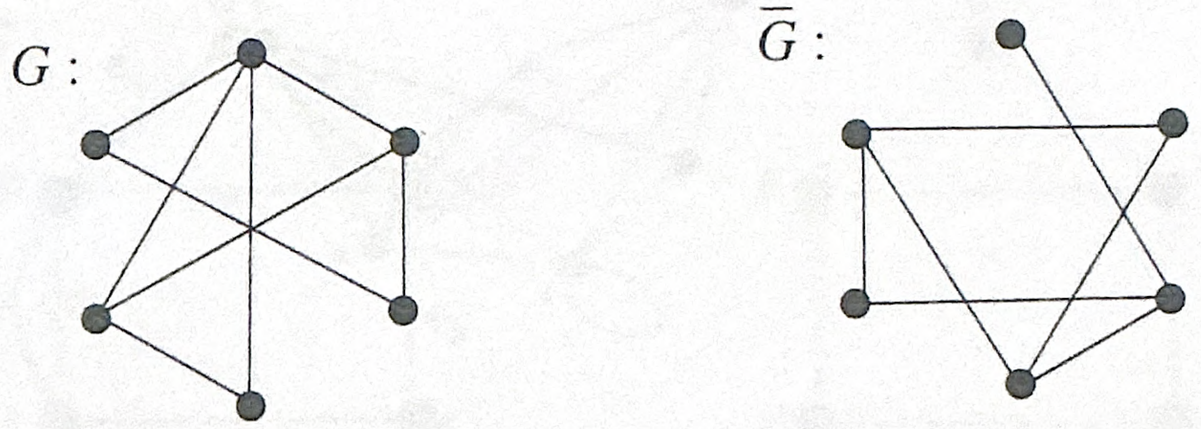
Definition: Let G=(V,E) be a simple graph. The ***complement*** of G is the simple graph $\overline{G} = (V, \overline{E})$ such that $\forall u, v \in V, (u \neq v) \Rightarrow (\{u, v\} \notin E \Leftrightarrow \{u, v\} \in \overline{E})$.

Note that the above FOL–wff is an abbreviation of:

$$(\forall u)(\forall v)(u \in V \wedge v \in V \wedge u \neq v \Rightarrow (\{u, v\} \notin E \Leftrightarrow \{u, v\} \in \overline{E})).$$

Example 8.6.11 In Figure 14, a graph G and its complement are displayed.

Figure 14



Definition: Let $G = (V, E, \varphi)$ and $G' = (V', E', \varphi')$ be two simple graphs such that $\varphi \cup \varphi'$ is a function. The ***union*** of G and G', is the graph $G \cup G' = (V \cup V', E \cup E', \varphi \cup \varphi')$.

When $V = V'$, $G \cup G'$ is the graph resulting from superimposing G' over G.

Lemma 8.6.6: *Let $G = (V, E)$ be a simple graph and $\overline{G} = (V, \overline{E})$ be the complement of G.*

(i) $E \cap \overline{E} = \emptyset$;

(ii) $G \cup \overline{G} = K_{|V|}$, *where* $K_{|V|} = (V, E_{K_{|V|}})$ *is the complete graph with vertex set* V.

Proof: (i) (Proof by contradiction)

Suppose $E \cap \overline{E} \neq \emptyset$. Then $(\exists x)x \in E \cap E$ (Lemma 3.5.8)

$\Rightarrow \{u, v\} \in E \cap \overline{E}$ (EI)

$\Rightarrow \{u, v\} \in E \wedge \{u, v\} \in \overline{E}$ (Definition of $\cap$)

$\Rightarrow \begin{cases} \{u, v\} \in E & \text{(I2)} \quad \cdots \text{(I)} \\ \{u, v\} \in \overline{E} & \text{(E9, I2)} \quad \cdots \text{(II)} \end{cases}$

$\Rightarrow u \in V \wedge v \in V \quad \cdots$ (III)

Since G is a simple graph, $u \neq v \cdots$ (IV)

$\Rightarrow u \in V \wedge v \in V \wedge u \neq v$ ((III),(IV),I6) $\cdots$ (IV)

It follows that:

$(\forall u)(\forall v)(u \in V \wedge v \in V \wedge u \neq v \Rightarrow (\{u,v\} \notin E \Leftrightarrow \{u,v\} \in \overline{E}))$ (Definition of $\overline{G}$)

$\Rightarrow u \in V \wedge v \in V \wedge u \neq v \Rightarrow (\{u,v\} \notin E \Leftrightarrow \{u,v\} \in \overline{E})$ (UI) $\cdots$ (V)

$\Rightarrow \{u,v\} \notin E \Leftrightarrow \{u,v\} \in \overline{E}$ ((IV),(V),I3)

$\Rightarrow (\{u,v\} \notin E \Rightarrow \{u,v\} \in \overline{E}) \wedge (\{u,v\} \in \overline{E} \Rightarrow \{u,v\} \notin E)$ (E20)

$\Rightarrow (\{u,v\} \in \overline{E} \Rightarrow \{u,v\} \notin E)$ (E9,I2) $\cdots$ (VI)

$\Rightarrow \{u,v\} \notin E$ ((II),(VI),I3) $\cdots$ (VII)

$\Rightarrow \{u,v\} \in E \wedge \{u,v\} \notin E$ ((I),(VII),I6)

$\Rightarrow$ a contradiction!

(ii) Since $G \cup \overline{G} = (V \cup V, E \cup \overline{E}) = (V, E \cup \overline{E})$ and $K_{|V|} = (V, E_{K_{|V|}})$,
to prove that $G \cup \overline{G} = K_{|V|}$, It suffices to prove that $E \cup \overline{E} = E_{K_{|V|}}$ or equivalently,
$\forall u, v \in V, u \neq v \Rightarrow \{u,v\} \in E \cup \overline{E}$.

(Direct proof)

Let $u, v \in V$ such that $u \neq v$.

Then $\{u,v\} \notin E \Leftrightarrow \{u,v\} \in \overline{E}$ (Definition of $\overline{G}$)

$\Rightarrow (\{u,v\} \notin E \Rightarrow \{u,v\} \in \overline{E}) \wedge (\{u,v\} \in \overline{E} \Rightarrow \{u,v\} \notin E)$ (E20)

$\Rightarrow \{u,v\} \notin E \Rightarrow \{u,v\} \in \overline{E}$ (I2)

$\Rightarrow\sim \{u,v\} \notin E \vee \{u,v\} \in \overline{E}$ (E18)

$\Rightarrow \{u,v\} \in E \vee \{u,v\} \in \overline{E}$ (E15)

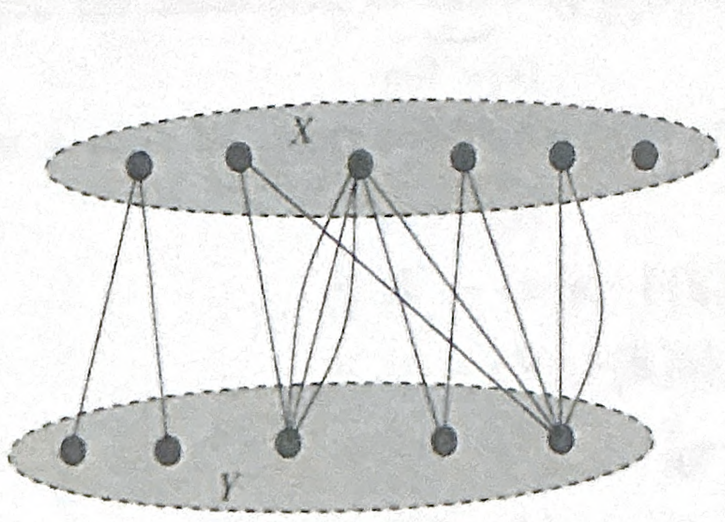
$\Rightarrow \{u,v\} \in E \cup \overline{E}$ (Definition of $\cup$)

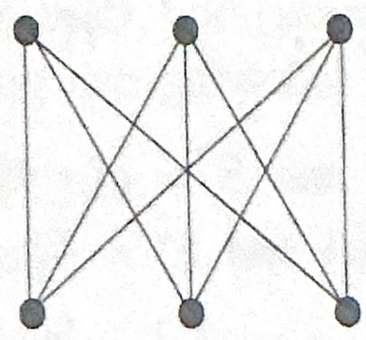
Therefore $E \cup \overline{E} = E_{K_{|V|}} \Rightarrow G \cup \overline{G} = K_{|V|}$, the complete graph with vertex set V. ■

8.7 Bipartite Graphs

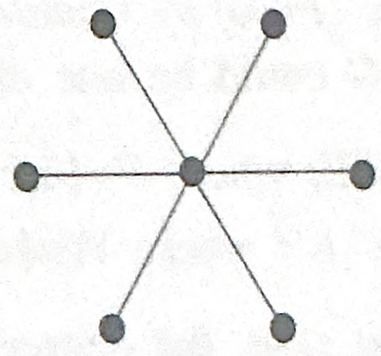
Definition: A graph $G = (V, E, \psi)$ is a ***bipartite graph*** if there exists $\{X, Y\}$ such that $X \cup Y = V$ and $X \cap Y = \emptyset$, and that every edge in the graph has one end-vertex in X and one end-vertex in Y. The set $\{X, Y\}$ is called a ***bipartition*** of the graph G. A ***complete bipartite graph*** is a simple bipartite graph with bipartition $\{X, Y\}$ such that every vertex in X is adjacent to every vertex in Y.

Let G_1, G_2 be any two complete bipartite graphs with bipartition $\{X, Y\}$ and $\{X', Y'\}$, respectively. Suppose $|X| = |X'| = m$ and $|Y| = |Y'| = n$. It is easily verified that the two graphs have identical graphical representation. They are thus isomorphic. It follows that, up to isomorphism, there is exactly one complete bipartite graph with bipartition of sizes m and n. We shall use $K_{m,n}$ to denote any such graph. Since $\{X, Y\}$ is a bipartition iff $\{Y, X\}$ is a bipartition, $K_{m,n} \cong K_{n,m}$. The graph $K_{1,m}$ is also called the ***star graph***.





$K_{3,3}$



$K_{1,6}$

8.8 Examples

1. Can a graph exist with degree sequence 5, 4, 4, 3, 3, and 2.

Solution: No! no graph can have an odd number of vertices with odd degree according to Theorem 8.4.2. ■

2. Show that a simple graph G cannot exist with degree sequence 3, 3, 3, 1.

Solution: (*Proof by Contradiction*) Suppose to the contrary that such a graph exists. Call the graph G. By removing the vertex of degree one and the edge incident upon it from G, we have a new graph $G'=(V',E')$ whose degree sequence is 3, 3, 2.

By Corollary 8.4.3, $|E'| \leq \frac{3(3-1)}{2} = 3$.
By Theorem 8.4.1, $|E'| = \frac{3+3+2}{2} = 4$.

We thus obtain $4 \leq 3$ which is a contradiction! Hence, G does not exist. ■

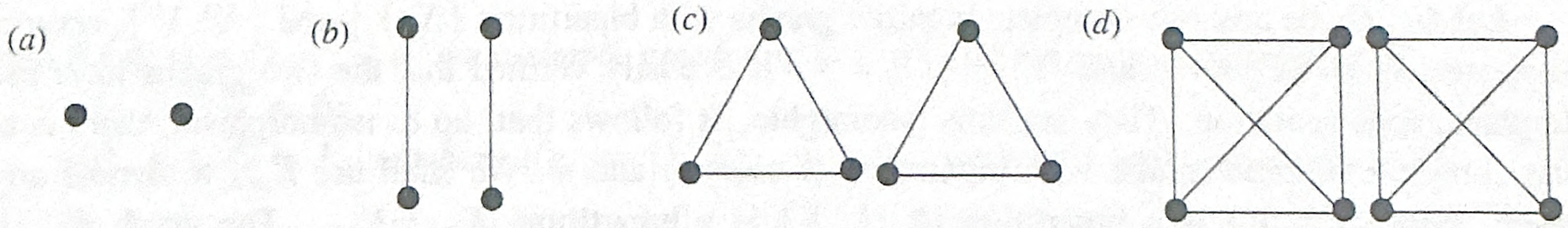
3. Give an example of:

(a) a 0-regular graph which is not a complete graph;
(b) a 1-regular graph which is not a complete graph;
(c) a 2-regular graph which is not a complete graph;
(d) a 3-regular graph which is not a complete graph;
(e) a k-regular graph which is not a complete graph.

Solution: (*Proof by Example*:)

(a) $G=(V,E)$ where $V=\{a,b\}$ and $E=\emptyset$. (Figure 15(a))
(b) $G=(V,E)$ where $V=\{a,b,c,d\}$ and $E=\{\{a,d\}, \{b,c\}\}$. (Figure 15(b))
(c) $G=(V,E)$ where $V=\{a,b,c,d,e,f\}$ and $E=\{\{a,b\}, \{b,c\}, \{c,a\}, \{d,e\}, \{e,f\}, \{f,e\}\}$. (Figure 15($c$))
(d) $G=(V,E)$ where $V=\{a,b,c,d,e,f,g,h\}$ and $E=\{\{a,b\}, \{b,c\}, \{c,d\}, \{d,a\}, \{a,c\}, \{b,d\}, \{e,f\}, \{f,g\}, \{g,h\}, \{h,e\}, \{e,g\}, \{f,h\}\}$. (Figure 15($d$))
(e) A graph consisting of two complete graphs of order $k+1$. ■

Figure 15



4. Suppose we know the order and size of a graph G. Is it possible to determine the degree sequence of G? Explain.

Solution: (*Proof by Counterexample*) No! Consider the case where the order of G is 4 and the size of G is 2. G could be one of the following two graphs:

$G=(V,E)$ where $V=\{a,b,c,d\}$ and $E = \{\{a,d\},\{c,d\}\}$, or
$G=(V',E')$ where $V'=\{a,b,c,d\}$ and $E' = \{\{a,c\},\{b,d\}\}$.

In the first case, the degree sequence of G is 2, 1, 1, 0.
In the second case, the degree sequence of G is 1, 1, 1, 1.

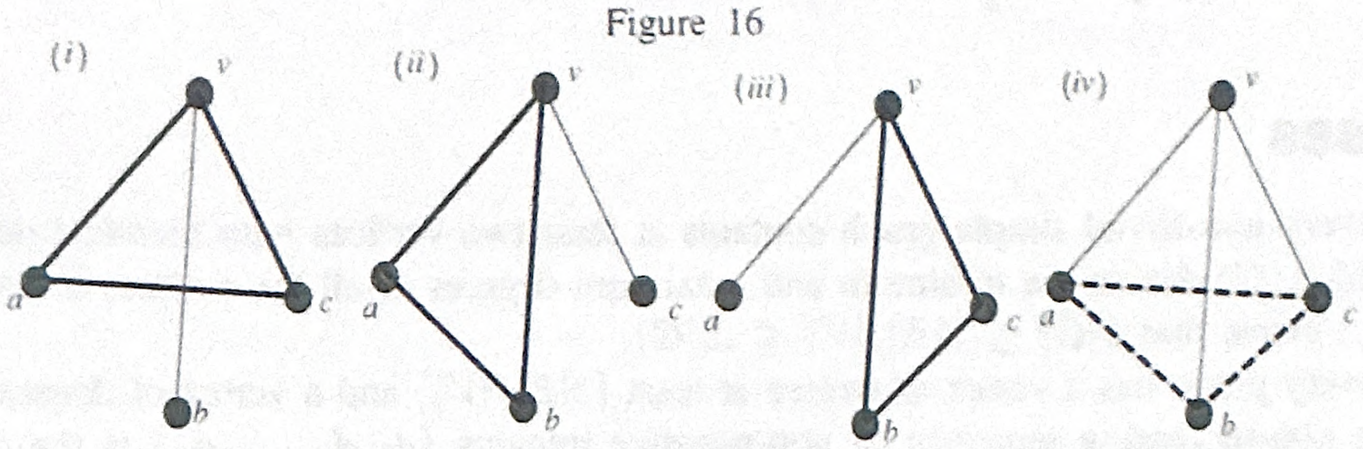
The two degree sequences are different. ■

5. Let G be a simple graph of order 6. Show that either G or $\overline{G}$ contains three mutually adjacent vertices.

Solution: Let v be any vertex in G. Since the order of G is 6, if $deg_G(v) < 3$, then $deg_{\overline{G}}(v) \geq 3$. i.e. v is adjacent to three other vertices in either G or $\overline{G}$. (Note: we can also use the Pigeonhole Principle. The pigeonholes are G and $\overline{G}$ and the pigeons are the 5 edges incident on v. By the Pigeonhole Principle, either G or $\overline{G}$ contains at least $\lceil \frac{5}{2} \rceil = 3$ of the 5 edges.)

Since the two cases are symmetrical, we need only consider the first case, i.e. that v is adjacent to three other vertices, a, b and c, in G. If any two of the vertices a, b, c are adjacent (Figure 16(*i*), (*ii*), (*iii*)),

Figure 16



then together with v, they form three mutually adjacent vertices in G. Otherwise, the three mutually non-adjacent vertices a, b, c are mutually adjacent in $\overline{G}$ (Figure 16(*iv*)). ■

6. Let G be a simple graph with order p and size q. Show that $q = \binom{p}{2}$ iff G is a complete graph.

Solution: $\Rightarrow$) (*Proof by Contradiction*) Suppose $q = \binom{p}{2}$ and G is not a complete graph. . . . . (I)

Since G is not a complete graph, there exist $u, w \in V$ such that $\{u, w\} \notin E$. . . . . (II)

As G is a simple graph, G has no self-loop.

$$\Rightarrow (\forall v \in V)\{v, v\} \notin E \quad \ldots\ldots \quad \text{(III)}$$

$$\Rightarrow \{u, u\} \notin E \quad \text{(UI)} \ \ldots\ldots \quad \text{(IV)}$$

It follows that $\{u, w\} \notin E \wedge \{u.u\} \notin E$ ((II),(IV),I6)

$$\Rightarrow deg(u) \leq p - 2 \ \ldots\ldots \quad \text{(V)} \quad \text{(Definition of } deg(u)\text{)}$$

Moreover, $\forall v \in V - \{u\}, \{v, v\} \notin E$ ((III),I3)

$$\Rightarrow \forall v \in V - \{u\}, deg(v) \leq (p - 1) \ \ldots\ldots \quad \text{(VI)} \quad \text{(Definition of } deg(v)\text{)}$$

By Theorem 8.4.1, $2q = \sum_{v \in V} deg(v)$

$$= \sum_{v \in V - \{u\}} deg(v) + deg(u) \qquad (\because V = (V - \{u\}) \cup \{u\})$$

$$\leq \sum_{v \in V - \{u\}} deg(v) + (p - 2) \qquad (\because \text{(V)})$$

$$\leq \sum_{v \in V - \{u\}} (p - 1) + (p - 2) \qquad (\because \text{(VI)})$$

$$\leq |V - \{u\}|(p - 1) + (p - 2)$$

$$\leq (p - 1)(p - 1) + (p - 2) \qquad (\because |V - \{u\}| = (p - 1))$$

$$< (p - 1)(p - 1) + (p - 1)$$

$$= p(p - 1)$$

$\Rightarrow q < \frac{p(p-1)}{2} = \binom{p}{2}$

$\Rightarrow q < q$, a contradiction! ((I), Sub$_=$)

$\Leftarrow$) (Direct proof) Suppose G is a complete graph. Then $deg(v) = (p-1), \forall v \in V$.

By Theorem 8.4.1, $2q = \sum_{v \in V} deg(v)$

$= \sum_{v \in V} (p-1)$ $(\because deg(v) = (p-1), \forall v \in V)$

$= p(p-1)$ $(\because |V| = p)$

$\Rightarrow q = \frac{p(p-1)}{2} = \binom{p}{2}$. ■

8.9 Exercises

1. Prove that every non-trivial simple graph contains at least two vertices with identical degrees.
2. Let $\delta(G)$ and $\Delta(G)$ denote the minimum and maximum degrees of all the vertices in an undirected graph G, respectively. Prove that $\delta(G) \leq 2|E|/|V| \leq \Delta(G)$.
3. Prove that every graph has a vertex of degree at least $\lceil 2|E|/|V| \rceil$ and a vertex of degree at most $\lfloor 2|E|/|V| \rfloor$.
4. Prove that a non-increasing sequence of non-negative integers $\{d_1, d_2, \cdots, d_n\}$ is the degree sequence of a graph (not necessary simple) iff $\sum_{i=1}^{n} d_i$ is even.
5. (a) Let G be a graph of order n and size m such that every vertex in it is of degree k or k+1. Prove that if G has N_k vertices of degree k and N_{k+1} vertices of degree k+1, then $N_k = (k+1)n - 2m$.
   (b) Let G be k-regular, where k is an odd integer. Prove that the size of G is a multiple of k.
6. Let G be a graph with 6 vertices and 10 edges in which every vertex has an odd degree. Suppose the number of vertices of degree 3 is one more than the number of vertices of degree 5. Determine the degree sequence of G.
7. Let G be a graph of order 14 and size 29 in which every vertex is of degree 3, 5, or 7. Suppose the number of vertices of degree 3 is twice the number of vertices of degree 5. Determine the degree sequence of G.
8. Let $G = (V, E)$ be a simple graph such that $v = \{u_1, v_1, u_2, v_2, u_3, v_3, u_4, v_4\}$ and $\{u_i, v_i\} \notin E, 1 \leq i \leq 4$. Suppose $deg(u) \neq deg(v), \forall u, v \in V - \{u_1\}$. Determine $deg(u_1)$ and $deg(v_1)$.
9. Prove that every graph $G = (V, E)$ without self-loop has a bipartite subgraph $G = (V', E')$ such that $|E'| \geq \frac{1}{2}|E|$.
10. Prove that the complete graph K_p can be expressed as the union of k bipartite graphs for all $p \leq 2^k$. [Hint: By induction on k.]
11. Let G be a simple graph having no isolated vertex and no induced subgraph consisting of exactly two edges. Prove that G is a complete graph.
12. Let $G = (V, E)$ be a k–regular simple graph with girth 4. Prove that $|V| \geq 2k$. Determine G if $|V| = 2k$.
13. Let $G = (V, E)$ be a simple graph with girth 5. If $deg(v) \geq k, \forall v \in V$, prove that $|V| \geq k^2 + 1$. Find one such graph G if $|V| = k^2 + 1$, for $k = 2$ and $k = 3$.